User Guide

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Based on our research paper “Hyperspectral Calibration Detection: A Novel Concept For Change Detection With Unsupervised Incremental Safe Pseudo-Labeling Implementation”, we prepare the source code for researchers to investigate our theory and algorithm. The prerequisites and the steps to run the algorithms are summarized below.

1 Prerequisites

The demo code is conducted under Ubuntu 22.04.3 LTS system with NVIDIA GeForce RTX 3090 GPU. The recommended software environment is listed below.

- Python 3.11.9
- PyTorch 2.1.0
- CUDA 12.2 (or newer)
- NumPy 2.3.5
- SciPy 1.16.3
- Scikit-learn 1.7.2

2 Dataset Preparation

The released demo currently supports the Yancheng/Farm dataset.

- **Farm1.mat**: first hyperspectral image
- **Farm2.mat**: second hyperspectral image
- **label.mat**: ground-truth change map (used only for evaluation)

Users may also add their own datasets by following the same format. Each dataset should contain two co-registered hyperspectral images of size $H \times W \times C$ and one binary ground-truth change map of size $H \times W$ for evaluation. The two hyperspectral images should have the same spatial and spectral dimensions and should be normalized to a comparable radiometric range. To use a new dataset, users need to add the corresponding data-loading branch in `load_data()` in `functions.py` and specify the dataset name in `config.yaml`.

3 Quick Start

Run “python main.py”. The program automatically performs

1. loading the hyperspectral images,
2. generating the initial safe pseudo labels,
3. training the calibration network,
4. updating the calibrated hyperspectral image,
5. refining the pseudo labels,
6. repeating the above procedure until convergence.

The calibrated model and the final change map will be automatically saved, and the corresponding evaluation metrics will be displayed in the command prompt.